

# At the eyepiece

What the Eye Still Knows. Thomas J. Watson Fellowship application, Aliaksandr Melnichenka, Berea College.

Everything on these two pages I made myself, with a phone held to an eyepiece or set on a wall. In Belarus the phone was a Redmi Note 7. The 2021 pictures were stacked in Sequator and adjusted in Lightroom. Page 1 is before Berea, page 2 is at Berea, and the people along the route follow on page 3.

## Before Berea. Mazyr and the family's country houses, 2020 to 2021.



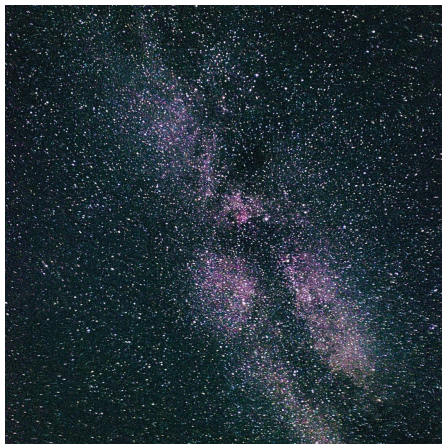
The Moon through my first telescope, the phone held to the eyepiece. Mazyr, August 2020.



The telescope that came the second New Year, the one that once showed me the neighbors' television across the yard, at the open window of my room on the fifth floor. Mazyr, June 2020.



One raw frame of the Milky Way, 32 seconds at ISO 3200, before stacking. Our country house, 10 August 2021.



Twenty-one such frames stacked into one picture. My friends asked whether it was Photoshop. August 2021.



Star trails and a meteor over the lake near my grandfather's house, stacked phone exposures. August 2021.



Fifty exposures of 32 seconds, nearly half an hour of sky, over my grandparents' house. August 2021.

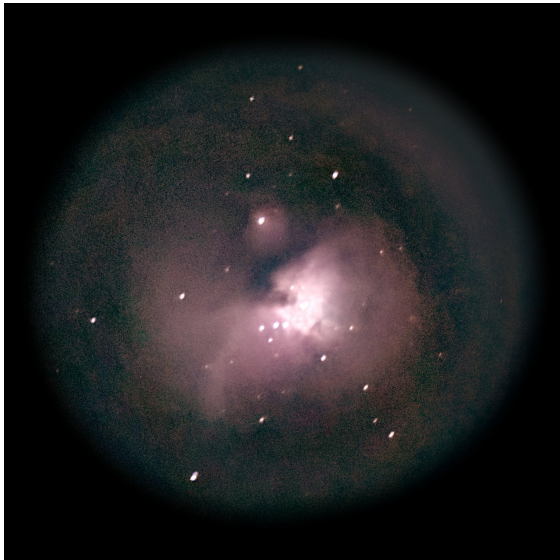
**At Berea. The physics department's 8-inch telescope, an iPhone held to the eyepiece, 2024 to 2026.**



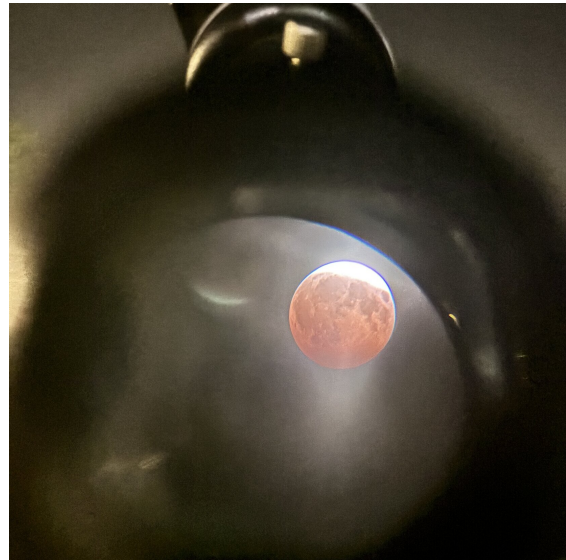
A visitor at the 8-inch on the observing deck of the science building during International Observe the Moon Night, which I helped set up. 14 September 2024.



The Moon through the 8-inch that night, the phone held to the eyepiece. 14 September 2024.



The Orion Nebula, M42, five phone frames at the eyepiece of the 8-inch, stacked, the night I first saw it, on a screen. Berea, 24 February 2025. One of the hundred Messier objects of the March marathon, so one percent done.



The partial lunar eclipse in the eyepiece during the Physics Club's public night on the alumni field. 27 August 2026.

# Hosts and correspondents

What the Eye Still Knows. Thomas J. Watson Fellowship application, Aliaksandr Melnichenka, Berea College.

I wrote to the observatories, societies and individual observers along the route of this year. The table lists the replies the year rests on, in the order I will meet the people, with the substance of what each one wrote. Words in quotation marks are theirs. The ten people in the application's project contacts are marked with an asterisk.

| Where and when                                                                | Who                                                                                                                                                                 | What they wrote                                                                                                                                                                                                                                                                                                                                             |
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| <b>Egypt, 22 July to about 20 August 2027. The Luxor eclipse of 2 August.</b> |                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                             |
| Kottamia and Luxor, July and August 2027                                      | <b>Dr. Bassem Nabawy*</b> President, National Research Institute of Astronomy and Geophysics (NRIAG), which runs the largest telescope in North Africa, at Kottamia | "You are most welcome to Katamiya on 26 July 2027." A hosted night at Kottamia on that date, at the institute's fee, and a place at its eclipse camps at Luxor, Qena or Marsa Alam once the camp fees are set, which he is still estimating "due to the very high rate of booking in Luxor."                                                                |
| Luxor, 2 August 2027                                                          | <b>Dr. Awni Al-Khasawneh*</b> Secretary General, Arab Union for Astronomy and Space Sciences                                                                        | Confirmed that the Union and its member societies will observe the eclipse at Luxor, that he will personally supervise them, and that I may coordinate my participation directly with him. He called it "your inspiring fellowship project" and wrote, "We are pleased to support your project."                                                            |
| Luxor, eclipse week                                                           | <b>Prof. Zainab Awad</b> Cairo University, co-chair of the university's eclipse conference in Luxor                                                                 | "You're most welcome to join the conference and attend the observations session." I may keep my own logbook during the session. She also asked how I had heard of the conference before its public announcement.                                                                                                                                            |
| Luxor, the evening of 2 August                                                | <b>Dr. Steve Bell</b> HM Nautical Almanac Office, UK Hydrographic Office                                                                                            | The Crescent MoonWatch project continues, "positive and negative sightings are always useful," and he listed what a report should carry, from the weather and the orientation of the horns to the observer's experience, where "First Time" is one of the boxes.                                                                                            |
| Luxor, the evening of 2 August                                                | <b>Prof. Bradley Schaefer</b> Louisiana State University                                                                                                            | "I am positively impressed with your idea of providing an outside observer for the Egyptian committee." He spent "a man-year of work" on the model of crescent visibility, explained why the false-alarm rate is the point of standing beside such a committee, and told me to expect little notice of my report. "Do what you think best, and act wisely." |

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| Where and when                 | Who                                                                                                                                                         | What they wrote                                                                                                                                                                                                                                                                                                   |
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| Luxor, the evening of 2 August | <b>Mohammad Odeh</b> Director, International Astronomical Center, Abu Dhabi, keeper of the world's crescent-sighting records, through Prof. Nidhal Guessoum | "A negative observation made by an experienced and reliable observer, under suitable and ideally good observing conditions, is no less important than a positive observation." It is the accumulated reports of not seeing that fix the limits of visibility. The evening beside the committee counts either way. |
| Luxor, 2 August 2027           | <b>Dr. Sarah Gibson</b> Solar physicist, High Altitude Observatory, NCAR, on the PUNCH mission team                                                         | "I will be at Luxor," as the scientist guide of an eclipse tour. In Texas in 2024 she watched the corona through polarized sunglasses and "could see a difference in the corona as I turned my head." "Luxor will be so long, there will be time to experiment!"                                                  |
| Luxor, 2 August 2027           | <b>Dr. Dale Frail</b> National Radio Astronomy Observatory, in Egypt as a tour's astronomer                                                                 | "My advice for people seeing their first eclipse is to not overthink it. Enjoy the moment and all its wonder." He went to his own first "with all the arrogance of a professional astronomer," and "it was awe inspiring."                                                                                        |
| Luxor, 2 August 2027           | <b>Allyson Bieryla</b> Harvard College Observatory, the LightSound project, which turns an eclipse into sound for blind listeners                           | Offers a pre-assembled LightSound device, or the parts to build one, for Luxor, and asks what I would do with it there.                                                                                                                                                                                           |
| Luxor, 2 August 2027           | <b>Dr. Hisashi Hayakawa</b> Nagoya University, who measures the Sun's radius from eclipse timings                                                           | "I'm looking forward to hearing about your measurements." He will stand on the same track in Morocco or Tunisia. Position to an arcsecond, time to a second, and "by naked-eye measurements, it's sometimes tricky to go beyond the precision of 1 s."                                                            |
| Wadi el Gemal, August 2027     | <b>Johannes Girardi</b> Gorgonia Beach, Wadi el Gemal National Park, Marsa Alam                                                                             | "Of course we are happy to support your experience." He will put me in touch with members of the Ababda community, the desert's own people, to guide the El Talayee trek, where my question for the guide is how they find their way by the stars, once we have spoken by call.                                   |
| The Eastern Desert             | <b>Prof. Joseph Hobbs</b> University of Missouri, who lived with the Khushmaan Bedouin of Egypt's Eastern Desert                                            | "Star lore was common wisdom among the Khushmaan in those days, and it had practical value." Al-Jadi, the North Star, for the way, and the heliacal rising of Sirius for the season.                                                                                                                              |

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| Where and when                              | Who                                                                                                                                         | What they wrote                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| Luxor, the first mornings of August 2027    | <b>Rumen Kolev</b> Author of The Babylonian Astrolabe, who kept a diary of the first morning of Sirius                                      | The morning Sirius first shows before dawn, which began the ancient Egyptian year, falls at Luxor between 31 July and 3 August 2027 by his tables, and on 30 July by Jeffrey Hunt's, so the eclipse week holds both. "Start your heliacal practice right now. Write down every session."                                                                                                                                                                                       |
| The Eastern Desert, the March 2028 new Moon | <b>Diaa Shawki</b> Desert and astronomy guide, Marsa Alam, recommended by the Oasis dive resort                                             | "It will be my pleasure to help." He wrote at eleven at night, back from two days guiding a group in the desert with a telescope, and will plan the new-Moon night for the Messier marathon. Desert permits come from the military, which is why a local guide is the way in.                                                                                                                                                                                                  |
| <b>Chile, late August to October 2027.</b>  |                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Elqui Valley, August 2027                   | <b>Gabriela López*</b> Head, Office for the Protection of the Quality of the Sky of Northern Chile (OPCC), with Juan Pablo Uchima           | "You can count on our support to accompany us during night sky brightness measurements." The round is made near the new Moon at a professional observatory, always, and the office measures every month whatever the winter does. Uchima added, "I assume you are used to cold weather, so that will probably not be the main issue."                                                                                                                                          |
| Vicuña, August to September 2027            | <b>Juan Daniel Durán Rojas</b> Office of the Mamalluca municipal observatory                                                                | "Su propuesta nos parece muy interesante." Willing in principle to have me alongside the observatory's guides and astronomers, the details to be coordinated at the start of 2027. "Esperamos poder recibirlo en Mamalluca," we hope to receive you.                                                                                                                                                                                                                           |
| Las Campanas, one night                     | <b>Dr. Juna Kollmeier*</b> Director, Sloan Digital Sky Survey, whose three-dimensional maps of the universe are the most detailed ever made | "Yes, this can be arranged." One night beside the survey's observers at Las Campanas. She added that my adviser Alex Lazarian once promised her he could read the turbulent field from the survey's Local Volume Mapper data alone, "maybe you will prove that he was telling the truth!" She applied for a Watson herself as a Caltech undergraduate with a similar plan, did not get it, went to Germany on a Fulbright instead, and offered the visit whatever the outcome. |
| Las Campanas, August to September 2027      | <b>Carol Rojas</b> Communications and outreach, Las Campanas Observatory, Carnegie Science                                                  | Answered with one question, the question of this whole year. "Would you rather talk to someone who knows the night sky by heart, rather than someone who works at an observatory and observes through a telescope?"                                                                                                                                                                                                                                                            |

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| Where and when                               | Who                                                                                                                                                        | What they wrote                                                                                                                                                                                                                                              |
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| La Silla, September 2027                     | <b>Dr. Petr Kabáth</b> Astronomical Institute of the Czech Academy of Sciences, the PLATOSpec spectrograph at the European Southern Observatory's La Silla | A day visit "should be possible" if his team is on the mountain, to be settled in March 2027. The telescope is run remotely from Europe or Santiago, and "there is no eyepiece for observing by eye during the night."                                       |
| La Silla and Paranal, Saturdays              | <b>Bárbara Núñez</b> Regional Relations Officer, European Southern Observatory, Santiago                                                                   | "You would be very welcome to join ESO's public Saturday tours at La Silla and Paranal," free and booked ahead, and a conversation in Vitacura in October 2027.                                                                                              |
| El Sauce, Río Hurtado                        | <b>Phil Evans</b> Amateur, exoplanet photometry by remote control from the Cook Islands with his telescope at El Sauce                                     | El Sauce is a farm of more than a hundred telescopes, most of them amateurs' and none with an eye at them. "I have not physically looked through an eyepiece in 20 years." He sent me on to Variable Stars South, whose Australian members still use theirs. |
| Santiago, if time allows                     | <b>Prof. Claudia Scóccola</b> Universidad de Chile, on the Simons Observatory team                                                                         | The QUBIC telescope she works with in the Argentine Andes is observed remotely, and nobody climbs to it. An open door at her university in Santiago whenever I am there.                                                                                     |
| Valparaíso, 13 to 20 October 2027            | <b>Domingo González*</b> President, Sociedad Astronómica de Valparaíso y Viña del Mar (SAVAL), an amateur society                                          | "It will be a pleasure to receive you, share experiences and learn more about your outstanding career." The society celebrates its seventieth anniversary in Valparaíso this October, "and we would love to have you."                                       |
| <b>Argentina, November to December 2027.</b> |                                                                                                                                                            |                                                                                                                                                                                                                                                              |
| La Plata and Buenos Aires, November 2027     | <b>Dr. Sixto Giménez Benítez</b> Archaeoastronomer, La Plata, suggested by César González-García of the European Society for Astronomy in Culture          | "Claro que puedo ayudarte," written from a congress in Rio de Janeiro. A La Plata visit in November 2027 should be no problem, help in Buenos Aires is easy, an afternoon on measuring a horizon marker's orientation by eye.                                |
| Buenos Aires, November 2027                  | <b>Dr. Cecilia Gómez</b> Anthropologist of the sky of the Gran Chaco peoples                                                                               | "I hope we can meet up in Buenos Aires so we can chat." She will bring the reading on how the Toba, Pilagá and Wichí read the sky, from the ethnographies of the 1920s to the journal Cosmovisiones.                                                         |

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| Where and when                          | Who                                                                                                                                                                   | What they wrote                                                                                                                                                                                                                                                                                                                                                                |
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| Buenos Aires, November 2027             | <b>Amigos de la Astronomía</b> One of the oldest amateur societies in Latin America. The association's secretary                                                      | Membership is open to a visitor, the student rate only with proof of current enrollment, so the ordinary rate for me. Members help with the association's activities, and with the guided visits once they know its history and its ways as well as the sky. "Recordá solicitar la baja cuando te vayas," remember to cancel the membership when you leave.                    |
| Barreal, 29 November to 8 December 2027 | <b>Dr. Antonio Damián Alejo*</b> Félix Aguilar Observatory, San Juan                                                                                                  | Is arranging visual observing nights for me at the observatory's Cesco station at Barreal, 2,350 meters up in the Andes. "Barreal is a very special place for astronomy, and I think your project fits very nicely with the astronomical tradition of the area."                                                                                                               |
| <b>Peru, January to February 2028.</b>  |                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                |
| Piura, 26 January 2028                  | <b>Dr. José del Carmen Silva Mechato*</b> Dean of the Faculty of Sciences, National University of Piura, with the directors of its schools of physics and mathematics | "Nos comprometemos," we commit, to carry out the observation of the annular eclipse with me on the Castilla campus, "an activity of great significance," with the coordination in the dean's own hands.                                                                                                                                                                        |
| Trujillo, 28 to 30 January 2028         | <b>Physics Student Center</b> National University of Trujillo, the local committee of the International Association of Physics Students                               | "We would be absolutely delighted to welcome you!" A place at one of their meetings, help with logistics, and a request that I give their students a talk, which I will.                                                                                                                                                                                                       |
| Chankillo, February 2028                | <b>Dr. Iván Ghezzi*</b> Director of Chankillo, the oldest solar observatory in the Americas                                                                           | "You are welcome to visit and conduct your research," with a formal letter first, standing a few meters from the western observing point from which people watched the Sun rise against thirteen towers around 300 BC. He advises the sunset, more certain than the sunrise in an El Niño year, and sent me to the table in his 2007 paper that gives the date for each tower. |
| Sacred Valley, 9 to 23 February 2028    | <b>Fernando Sánchez Montero</b> Sembrando Cielos, Calca                                                                                                               | "In principle I would be happy to have you join us for a night." Help alongside his team and the Andean dark constellations by eye, Yacana the llama among them. "And if the weather does not allow us to observe the sky, you can still count on our support," an indoor session at his visitor space. Permission to name him given.                                          |

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| Where and when                                                        | Who                                                                                                                                                    | What they wrote                                                                                                                                                                                                                                                                                                                                                          |
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| <b>Australia, March to May 2028, and the eclipse of 22 July 2028.</b> |                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                          |
| New South Wales, autumn 2028                                          | <b>Prof. Ray Norris*</b> CSIRO Space and Astronomy, and Western Sydney University, a radio astronomer and a leading researcher of Aboriginal astronomy | Will put me in touch with a Gamilaraay elder once the elder gives permission. On the Emu in the Sky, “most people interested in this area will already be familiar with the shape of the emu in the sky, just from looking at the sky, but it may be from conversations with a knowledge holder that they learn about the subtleties of how it varies through the year.” |
| Dubbo, late April and 22 July 2028                                    | <b>Peter Starr</b> Dubbo Observatory, New South Wales                                                                                                  | “Yes we are planning to run a session for this and have some telescopes set up for it.” He offered to put me on the list for eclipse day, inland from Sydney’s winter cloud, and the observatory shows the sky through telescopes every night.                                                                                                                           |
| South Australia, if I detour                                          | <b>Andrew Cool</b> River Murray International Dark Sky Reserve                                                                                         | Has measured the reserve’s sky brightness since 2018. “At our latitude, Sagittarius passes overhead,” and the new observatory at Swan Reach has a telescope “big enough to allow colour to be perceived in the Trifid nebula through the eyepiece.”                                                                                                                      |
| Sydney, 22 July 2028                                                  | <b>Meredith Hall</b> Manager, Inspiring Australia (NSW), the national science engagement program, University of Wollongong                             | “Wonderful to hear from you and the Fellowship sounds like an exciting opportunity. We would be glad to assist with finding you a volunteer opportunity to be a part of the eclipse celebrations.”                                                                                                                                                                       |
| Sydney, autumn 2028                                                   | <b>Hans Feitz</b> President, Sutherland Astronomical Society, Green Point Observatory                                                                  | The observatory was built almost sixty years ago “when the skies were dark and the trees were small,” and now teaches its members to use their telescopes, with an eight-week course and open nights. “In our Bortle 6 sky we can just make out 4 of the 5 Crux stars and only during a new moon, all 5.”                                                                |
| Coonabarabran, April 2028                                             | <b>New South Wales Department of Education</b> STEM team, who introduced me to Wes Leedham of the Warrumbungle Environmental Education Centre          | A school or teacher near Coonabarabran who might take a visiting student into an astronomy lesson, the introduction made, the answer to come.                                                                                                                                                                                                                            |

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| Where and when                             | Who                                                                                                                                | What they wrote                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| Dark Sky Hill, 6 to 12 April 2028          | <b>Mark and Wendy Osmond</b><br>Dark Sky Eco, a stay on Gamilaraay country, New South Wales                                        | “Guests still put an eye to glass. The night is built in three layers, and they are not substitutes for each other. By eye first. No telescope. That is where the Cycle of Creation starts, and it is where the Emu lives.” At the eyepiece guests say what they never say at the screen, “That’s actually there.” “Is that light hitting my eye?” “The screen can show you more structure. It cannot give you that particular shock.” Book like any guest and the skywatch is part of the stay if the weather gives a night. |
| Broken Hill                                | <b>Trevor Barry</b> Amateur planetary observer, co-author of the Science Advances paper on the decagon wave at Saturn’s south pole | The other side of the argument, from an amateur with a world-first discovery. “I have never had an eyepiece in this telescope, all my data is avi video at 72 fps.”                                                                                                                                                                                                                                                                                                                                                           |
| <b>New Zealand, May to mid July 2028.</b>  |                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Kaikōura, 8 to 21 May 2028                 | <b>Lisa Bond*</b> Kaikōura Dark Sky Trust                                                                                          | “You would certainly be very welcome to visit us.” A round of the sanctuary’s sky-quality readings with the team, and any community stargazing evening or school session in my two weeks. “Being in the Southern Hemisphere will also give you a wonderful opportunity to experience objects in our night sky that you won’t see from your part of the world.” Permission to name the Trust given.                                                                                                                            |
| Christchurch, May 2028                     | <b>Hubertien Wichers</b> President, New Zealand Antarctic Society                                                                  | Where the recorded voices of the men who watched the southern sky from the ice in the 1950s are kept, and how to hear them. “Please contact me when you visit, also if you are looking for accommodation. I’d love to catch up and perhaps you would like to present your research so far.”                                                                                                                                                                                                                                   |
| Wairarapa and Wellington, May to June 2028 | <b>Dr. Haritina Mogoşanu</b> Star Safari observatory and Spaceward Bound New Zealand                                               | “Fantastic to hear from you and great credentials.” Introductions to the Victoria University physics team and to teachers across the country, her observatory “in the best darkest place in New Zealand,” the national astronomy olympiad team she runs, and an offer of a letter. “Yes you can say that.”                                                                                                                                                                                                                    |
| Wellington, June 2028                      | <b>Dr. Yvette Perrott</b> Victoria University of Wellington, who runs New Zealand’s astronomy olympiad selection                   | The selection exam has no observing task yet. “It may be worth thinking about putting a question or two in there, and/or having a training evening for the team if we’ve selected them by the time you’re here. Let’s keep in touch.”                                                                                                                                                                                                                                                                                         |

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| Where and when                   | Who                                                                                            | What they wrote                                                                                                                                                                                                                                                                                                                   |
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| Wellington, May to July 2028     | <b>Adam Rosner</b> Wellington Astronomical Society                                             | The society's meetings are public. Star Field nights at its dark-sky site on 20 or 27 May, 17 or 24 June and 22 July 2028, with a ride shared over the mountains. Camping there in June means nights below freezing, "Translation: It gets cold at night," so he pointed me to the town of Martinborough.                         |
| Wellington, May 2028             | <b>Briar</b> Visitor Experience Lead, Space Place, Wellington                                  | "We'd be happy to have you along to observe and support with operating the telescopes on one of those nights." The staff always find the target for the visitor, and "sometimes young children find it hard to see through the eyepiece."                                                                                         |
| Naseby, June to July 2028        | <b>Paul Bishop</b> Night-sky tours, Naseby, Central Otago                                      | Almost every evening of June and July 2028 free, the only exception being the eclipse itself, and a 12-inch telescope that can be driven by hand. "That sounds perfect!"                                                                                                                                                          |
| Lauder, June 2028                | <b>Richard Querel</b> Lauder atmospheric research station, Earth Sciences New Zealand          | "Yes, we host visitors and can give a tour." A chance to watch an ozone balloon launch if the weather is calm, usually on a Monday or Tuesday. The balloons are two meters across when filled, followed by eye until they are a dot about a kilometer up, and burst near 34 kilometers. "Your BUSEL-1 project looks interesting." |
| Lower South Island, June 2028    | <b>Ian Cooper</b> Aurora observer by eye since 1978, Royal Astronomical Society of New Zealand | Sent the society's visual report form, which he updated in 2024, and how to write a sighting down. "The eye could see fine detail that was often blurred on film." Auroras that only a camera picks up "we call subvisual," and he uses his own camera just to confirm a glow he suspects.                                        |
| Invercargill, June 2028          | <b>Elizabeth King</b> President, Southland Astronomical Society                                | Welcome at the society's monthly meeting and its public viewing night. She has chased the aurora for fifteen years, green on strong nights and red and pink on the best. "What I see with the naked eye others may not."                                                                                                          |
| Dunedin, 26 June to 15 July 2028 | <b>Jessica Hatfield and Georgette Watkins</b> Tūhura Otago Museum                              | Volunteer opportunities in and around the planetarium. "Happy for you to mention we have opportunities available for you."                                                                                                                                                                                                        |
| Dunedin, 14 July 2028            | <b>Michelle Taiaroa</b> University of Otago, Kāi Tahu                                          | The Matariki dawn ceremony on the museum lawn, and how to attend it respectfully. "During the ceremony you will start to hear people call out names. They are the names of their loved ones that have passed away since the last Matariki." Breakfast in the museum afterwards.                                                   |

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| Where and when                             | Who                                                                                                                  | What they wrote                                                                                                                                                                                                                                                                                 |
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| <b>Advisers on the eye, for every leg.</b> |                                                                                                                      |                                                                                                                                                                                                                                                                                                 |
| Anywhere                                   | <b>Stephen O'Meara</b> Discovered the spokes in Saturn's rings by eye                                                | How an eye trained to a tenth of a magnitude on variable stars found the spokes through a 9-inch refractor at Harvard. He tried for a Moon a few hours old in Mallorca this year and lost it in the smoke of mainland fires. "But you can always try."                                          |
| Anywhere                                   | <b>Prof. Alex Filippenko</b> University of California, Berkeley, twenty-one totalities                               | "What I do emphasize is NOT to let equipment or tasks dominate the diamond rings and totality." Cameras and stopwatches "are distractions that keep you from absorbing the phenomenon as it unfolds," and the preparation he gives first-timers has never, they tell him, diminished the sight. |
| Anywhere                                   | <b>Prof. Ralph Chou</b> University of Waterloo, the optometrist behind the eclipse eye-safety standards              | How to reach totality dark-adapted, glasses on until only a couple of Bailey's beads remain, then "plenty of time to recover dark adaptation and see the corona in its glory." He will be in Morocco on the same track, and warns that three hours in the Egyptian sun is dangerous.            |
| Anywhere                                   | <b>Dr. Michele Bianda</b> Specola Solare Ticinese, Locarno, the reference observatory for the world's sunspot number | Why the sunspot number is still counted "by eye, pencil and paper sheet." The series runs from Galileo's drawings, satellites are calibrated against it, and keeping the method human keeps four centuries comparable. "Before to stop this series it is worthwhile to think with quiet."       |
| Anywhere                                   | <b>Dr. Laure Lefèvre</b> Royal Observatory of Belgium, the SILSO network that produces the world's sunspot number    | How a network of human counters catches one observer drifting, and why a station ends. "An ending often looks at first exactly like a gap." The observer "becomes older, has health or eyesight problems, moves," and sometimes contact is simply lost.                                         |
| Anywhere                                   | <b>Dr. Benjamin Boe</b> Wentworth Institute of Technology, who images the corona at eclipses                         | "It is remarkable how much you can see with just your eyes, and how it far surpasses a single camera exposure." The cameras win only by stacking many exposures, and he has never seen shadow bands himself, "always been too focused."                                                         |
| Anywhere                                   | <b>Gary Poyner</b> British Astronomical Association, 370,000 visual estimates                                        | Why he stayed at the eyepiece when cameras arrived. "There is something very special in seeing a rare outburst of a dwarf nova, sometimes for the very first time." On a clear night he covers more than 250 stars by eye.                                                                      |
| Anywhere                                   | <b>Joanna DiStefano</b> American Association of Variable Star Observers                                              | Associate membership and the mentor program for learning visual estimates. Her mother went to Berea, and she herself once had dinner at Boone Tavern with Loyal Jones of the Appalachian Center.                                                                                                |

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| Where and when | Who                                                                                                                   | What they wrote                                                                                                                                                                                                                                                                                             |
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| Anywhere       | <b>Dr. Mitsuru Sôma</b> Vice President for Grazing Occultation Services, International Occultation Timing Association | “Observations do not need a team. A beginner also can make a graze report.” Timings called by voice onto a recorder with a time signal under it are still useful, and “I look forward to your first graze report.”                                                                                          |
| Anywhere       | <b>Prof. Trevor Lamb</b> Australian National University, the physiology of dark adaptation                            | How long the rods take to recover after a glance at a screen, “possibly 10 mins,” and why. “Do NOT use a phone/laptop if you want your best scotopic threshold.” A deep-red filter if I must.                                                                                                               |
| Anywhere       | <b>Ian Ridpath</b> Author of Star Tales, editor of the Oxford Dictionary of Astronomy                                 | How navigators found the south celestial pole by eye, down the long axis of the Southern Cross to where it meets the perpendicular from Alpha and Beta Centauri. “Amateur astronomers still use the same technique today.”                                                                                  |
| Anywhere       | <b>Mike Frost</b> Eclipse chaser, fourteen totalities and counting, British Astronomical Association                  | What is different about the tenth one. “Totality doesn’t last long. My first total eclipse lasted 45 seconds. It leaves you wanting more.” At second contact there are five places to look at once, and “you can’t do all these things simultaneously. Solution, go see another eclipse.”                   |
| Anywhere       | <b>Christine Malec</b> Blind writer, co-host of the podcast Talk Description to Me                                    | How to describe the Moon to someone who cannot see it. “Your sighted audience will be able to look at images on a screen and know the visual details. What they won’t know is what it’s like for someone from the northern hemisphere to see a night sky they’ve never seen before.”                        |
| Anywhere       | <b>Dr. Christopher Kyba</b> Light-pollution scientist, Ruhr University Bochum                                         | A trained eye can judge its limiting magnitude to a tenth, but the sky changes more than that from night to night, so “one observer (person or machine) at one site cannot observe a trend until several years have gone by.” The worth of my readings is in joining thousands of others at Globe at Night. |
| Anywhere       | <b>Prof. Matthew Bailes</b> Swinburne University, co-discoverer of the first fast radio burst                         | How the first burst was found at Parkes, a threshold in software and then a person looking at a plot. Today’s searches run about a billion trials a second. “You can’t keep the raw data.”                                                                                                                  |
| Anywhere       | <b>Alina Sabyr</b> 2019 Watson Fellow                                                                                 | “Your project sounds really fascinating.” A Zoom call on a year that starts with an eclipse, once her busy weeks pass.                                                                                                                                                                                      |